

V.F.S. v2.0 (Open-Gate)

Integrated Cosmological Layer v8

The Brim as scale factor, Sophia as modal source, Folding as stabilizing form

Derived companion to `vfs_geometry.tex` and `vfs_opengate_core.tex`
Integrated with the Sophianic Folding module

Scope and epistemic status

The geometric layer treats the Brim Ω_P as the time of a conformal flow of the 2D live surface. This file takes the Lorentzian lift: the Brim is the **timelike** axis of expansion, and the live surface is a spatial slice. The result is an open (2+1)-dimensional Friedmann–Lemaître–Robertson–Walker (FLRW) spacetime whose scale factor is the Brim. *Propositions* are closed-form consequences; *Corollaries* are interpretive readings. Three standing caveats carry over:

- The expansion direction is the flow/time axis, lifted with $-dt^2$; it is **not** a Riemannian radial coordinate. Promoting Ω_P to a spacelike coordinate (a metric cone) misassigns its dynamical role.
- The spatial metric is the canonically warped hyperbolic live surface (geometric layer), fixed only up to that warp.
- V.F.S. supplies the scale-factor law $\dot{\Omega}_P = \alpha\lambda$; the “effective content” below is the stress-energy that General Relativity *would* require to source this expansion — a reconstruction, not an independent postulate of gravity.

Integrated folding convention. In this rewritten version Sophianic Folding is not presented as a later historical patch. It is built into the cosmological dictionary from the start. The symbol q is therefore reserved no longer for two different roles: the deceleration parameter is written q_{dec} , while the morphological folding amplitude is q_{fold} and its non-negative intensity is

$$Q_{\text{fold}} = q_{\text{fold}}^2.$$

Thus the cosmological question “does the Vessel accelerate or decelerate?” and the geometric question “does the Vessel change form?” remain distinct, but coupled.

Dimensional principle. The model is *canonically* 2+1: a two-dimensional Vessel-surface whose scale factor is the Brim $\Omega_P(t)$. The Brim-expansion principle is complete in this setting, and 2+1 is apt rather than merely sufficient — in $D \leq 3$ the Weyl tensor vanishes, so the spacetime curvature is fixed pointwise by the source (Sophia) with no free gravitational remainder. Any three-dimensional rendering is a *visualization only*: its radial direction is an embedding artifact with no V.F.S.

meaning, and the static Riemannian cone $d\Omega_P^2 + \Omega_P^2 h$ is a different object from this spacetime (its scalar $-4/\Omega_P^2$ is not the spacetime curvature). The sole legitimate third *spatial* coordinate would be a genuine state variable (σ or λ), belonging to the geometry of phase space — a separate question from Brim-flow.

1 From Live-Surface Flow to Vessel Spacetime

The geometric layer gives, on each slice of fixed Brim, the hyperbolic live surface $h = dy^2 + e^{-2y} d\Delta^2$ with $y = -\ln(1-r)$, $r = u/\Omega_P$, of constant curvature $K_h = -1$. The Brim-flow $\partial_t g_\Sigma = 2(\alpha\lambda/\Omega_P)g_\Sigma$ is a motion in time, so its Lorentzian home assigns the expansion to a timelike direction:

$$ds_{\text{Vessel}}^2 = -dt^2 + \Omega_P(t)^2 h, \quad a(t) \equiv \Omega_P(t).$$

This is the open ($k = -1$) FLRW background in 2+1 dimensions, with scale factor equal to the Brim. In the integrated Open-Gate version the spatial slice may also undergo a first Sophianic shape-mode:

$$g_\Sigma^{\text{fold}}(t) = \Omega_P(t)^2 (h_0 + q_{\text{fold}}(t)e_1), \quad Q_{\text{fold}} = q_{\text{fold}}^2.$$

The pure FLRW metric is the background branch $q_{\text{fold}} = 0$; the folded branch keeps the same Brim scale factor but gives the Vessel a new spatial form. *The “expanding inner universe” is genuine, but it expands in time: Ω_P is the scale factor, not a spatial radius. Sophianic Folding does not replace Brim expansion; it adds a stabilizing shape-response on the spatial slices.*

2 Curvature of the Vessel Spacetime

Proposition 1 (Spacetime curvature). *For $ds_{\text{Vessel}}^2 = -dt^2 + a^2 h$ with $K_h = -1$ in 2+1 dimensions,*

$$R = \frac{2(2a\ddot{a} + \dot{a}^2 - 1)}{a^2} = \frac{2(2\Omega_P \alpha \dot{\lambda} + \alpha^2 \lambda^2 - 1)}{\Omega_P^2}.$$

For sub-exponential expansion ($\ddot{a} \rightarrow 0$, $\dot{a}$ bounded) $R \rightarrow 0$: the spacetime flattens. For de Sitter expansion ($a = e^{Ht}$) $R \rightarrow 6H^2$, a constant: maximal inflation does not flatten (§13). The folded spatial slice carries an additional curvature contribution

$$R_{\text{spatial}}^{\text{fold}} = -\frac{2}{\Omega_P^2} + \rho_1 Q_{\text{fold}}.$$

Thus global Brim expansion and local curvature concentration can coexist.

3 Kinematics: Hubble, Acceleration, and Folding Shadow

Proposition 2 (Hubble and deceleration). *With $a = \Omega_P$ and $\dot{\Omega}_P = \alpha\lambda$,*

$$H = \frac{\dot{a}}{a} = \frac{\alpha\lambda}{\Omega_P}, \quad q_{\text{dec}} = -\frac{\ddot{a} a}{\dot{a}^2} = -\frac{\dot{\lambda} \Omega_P}{\alpha \lambda^2}, \quad \text{sign } q_{\text{dec}} = -\text{sign } \dot{\lambda}.$$

The Hubble rate of the Vessel is exactly the conformal Brim-flow rate $\psi = \alpha\lambda/\Omega_P$. Expansion accelerates ($q_{\text{dec}} < 0$) if and only if Sophia is growing: accelerated Theosis is growing wisdom.

In the Open-Gate folding branch, the Sophia flow is regulated by the morphology:

$$\dot{\lambda}_{\text{fold}} = J_{\text{in}} - \gamma\lambda - \eta Q_{\text{fold}}\lambda, \quad J_{\text{in}} = k_\sigma\sigma + I_{\text{gate}}.$$

Hence the folded deceleration parameter is

$$q_{\text{dec}}^{\text{fold}} = q_{\text{dec}}^{(0)} + \frac{\eta\Omega_P}{\alpha\lambda} Q_{\text{fold}}, \quad q_{\text{dec}}^{(0)} = -\frac{\Omega_P}{\alpha\lambda^2} (J_{\text{in}} - \gamma\lambda).$$

This is the cleanest cosmological meaning of folding: q_{dec} is not identical to q_{fold} , but the observable stabilization of the expansion is the cosmological shadow of Q_{fold} .

4 Effective Content: Sophia as Cosmological Source

Treating the reconstructed stress-energy as a perfect fluid ($8\pi G = 1$, comoving frame), the Einstein tensor gives:

Proposition 3 (Effective Friedmann content). *On the background branch $Q_{\text{fold}} = 0$,*

$$\rho_{\text{eff}} = \frac{\alpha^2\lambda^2 - 1}{\Omega_P^2} = H^2 - \frac{1}{\Omega_P^2}, \quad p_{\text{eff}} = -\frac{\ddot{a}}{a} = -\frac{\alpha\dot{\lambda}}{\Omega_P}, \quad w = \frac{p_{\text{eff}}}{\rho_{\text{eff}}} = \frac{-\alpha\dot{\lambda}\Omega_P}{\alpha^2\lambda^2 - 1}.$$

On the folded branch,

$$p_{\text{eff}}^{\text{fold}} = -\frac{\alpha\dot{\lambda}_{\text{fold}}}{\Omega_P}, \quad \rho_{\text{eff}}^{\text{fold}} = \frac{\alpha^2\lambda^2 - 1}{\Omega_P^2} + \frac{\rho_1}{2} Q_{\text{fold}},$$

$$w_{\text{fold}} = \frac{-\alpha\dot{\lambda}_{\text{fold}}\Omega_P}{\alpha^2\lambda^2 - 1 + (\rho_1/2)\Omega_P^2 Q_{\text{fold}}}.$$

Sophia is the cosmological source, and its rate $\dot{\lambda}$ sets the pressure. Folding does not abolish expansion; it regulates the rate of Sophia growth and contributes a curvature-density correction through Q_{fold} .

5 Sophianic Folding as Cosmological Stabilization

The folding intensity is inherited from the first shape-mode of the Vessel:

$$Q_{\text{fold}} = \frac{A_{\text{VFS}}}{B_{\text{VFS}}}, \quad A_{\text{VFS}} = (c_0 - c_1\gamma)\lambda + c_1k_\sigma\sigma + c_1I_{\text{gate}} - a_0, \quad B_{\text{VFS}} = b + 2c_1\eta\lambda.$$

The regular domain is

$$0 < Q_{\text{fold}} < q_{\text{max}}^2.$$

In that domain folding is not a singular event. It is the morphodynamic mechanism by which excess Open-Gate Sophia becomes form and thereby stabilizes the cosmological acceleration. The stabilizing contribution to deceleration is

$$\Delta q_{\text{dec}} = \frac{\eta \Omega_P}{\alpha \lambda} Q_{\text{fold}} > 0.$$

Thus Open-Gate can still sustain expansion, while folding prevents unlimited acceleration from being the only possible meaning of Sophia growth.

The same process may resolve morphological tension. This is not physical gravitational energy; it is a Lyapunov/shape potential. If

$$\mathcal{T}_{\text{fold}}(q_{\text{fold}}) = -\frac{1}{2} A_{\text{VFS}} q_{\text{fold}}^2 + \frac{1}{4} B_{\text{VFS}} q_{\text{fold}}^4,$$

then along the shape flow

$$J_{\text{rel}} = -\dot{\mathcal{T}}_{\text{fold}} = \dot{q}_{\text{fold}}^2 \geq 0.$$

A symbolic Pleromatic output channel may therefore be written as

$$\dot{P} = \dot{P}_{\text{core}} + \kappa_{\text{pler}} \theta_{\text{pler}} J_{\text{rel}},$$

without interpreting J_{rel} as ordinary gravitational stress-energy.

6 Sophia is Modal, Not Typological

A frequent mistake is to treat the cosmological regimes below as different kinds of Sophia. In this model Sophia is not typological but modal. There is one symbolic quantity,

$$\lambda(t)$$

whose accumulated value, growth rate, and folding embodiment appear differently in the cosmological reconstruction. Thus the same Sophia can appear as density, pressure, acceleration, or form depending on which derived observable is being read.

The modal dictionary is:

λ	$\mapsto$	Brim expansion and effective density,
$\dot{\lambda}_{\text{fold}}$	$\mapsto$	pressure and acceleration,
Q_{fold}	$\mapsto$	morphological stabilization and curvature correction,
$J_{\text{rel}} = -\dot{\mathcal{T}}_{\text{fold}}$	$\mapsto$	Pleromatic release / realized output.

In words:

Accumulated Sophia expands; growing Sophia accelerates; folded Sophia stabilizes.

Folding is therefore not a new kind of Sophia. It is one mode by which Sophia becomes form.

7 Zero-Density Threshold and its Folding Shift

On the background branch the reconstructed effective density changes sign at

$$\rho_{\text{eff}} = 0 \iff \alpha\lambda = 1.$$

This gives the classical Open-Gate threshold:

Condition	ρ_{eff}	Cosmological reading
$\alpha\lambda < 1$	$\rho_{\text{eff}} < 0$	sub-vacuum / below Milne content
$\alpha\lambda = 1$	$\rho_{\text{eff}} = 0$	Milne threshold
$\alpha\lambda > 1$	$\rho_{\text{eff}} > 0$	positive effective content

With Sophianic Folding the density is instead

$$\rho_{\text{eff}}^{\text{fold}} = \frac{\alpha^2\lambda^2 - 1}{\Omega_P^2} + \frac{\rho_1}{2}Q_{\text{fold}}.$$

Hence the folded zero-density locus becomes

$$\rho_{\text{eff}}^{\text{fold}} = 0 \iff \alpha^2\lambda^2 = 1 - \frac{\rho_1}{2}\Omega_P^2Q_{\text{fold}}.$$

Therefore regular folding lowers the threshold at which Sophia becomes positive effective content, provided $\rho_1 Q_{\text{fold}} > 0$. In the background model Sophia must reach $\alpha\lambda = 1$; in the folded model part of the burden is carried by the curvature-form term.

Corollary 1 (Folding lowers the positive-content threshold). *If $\rho_1 > 0$ and $Q_{\text{fold}} > 0$, then positive effective content can arise before the background Milne threshold $\alpha\lambda = 1$. Sophianic Folding therefore allows form to contribute to the cosmological readability of Sophia as content.*

8 Pressure-Axis of Sophia

The pressure axis is controlled not by the accumulated amount of Sophia alone, but by its rate of change. In the background branch,

$$p_{\text{eff}} = -\frac{\alpha\dot{\lambda}}{\Omega_P}.$$

Thus:

Sophia rate	p_{eff}	Cosmological mode
$\dot{\lambda} > 0$	$p_{\text{eff}} < 0$	negative pressure / acceleration
$\dot{\lambda} = 0$	$p_{\text{eff}} = 0$	coasting / dust-like limit
$\dot{\lambda} < 0$	$p_{\text{eff}} > 0$	positive pressure / deceleration

In the folded branch the relevant rate is

$$\dot{\lambda}_{\text{fold}} = J_{\text{in}} - \gamma\lambda - \eta Q_{\text{fold}}\lambda.$$

Consequently,

$$p_{\text{eff}}^{\text{fold}} = -\frac{\alpha \dot{\lambda}_{\text{fold}}}{\Omega_P}.$$

The term $-\eta Q_{\text{fold}} \lambda$ is the cosmological appearance of morphological stabilization: it reduces the pressure-producing growth of Sophia without reversing Brim expansion itself.

9 Folded Density-Pressure Plane

The old modal plane was

$$(\lambda, \dot{\lambda}) \mapsto (\rho_{\text{eff}}, p_{\text{eff}}, w).$$

The integrated Open-Gate version is the folded density-pressure-form space:

$$(\lambda, \dot{\lambda}_{\text{fold}}, Q_{\text{fold}}) \mapsto (\rho_{\text{eff}}^{\text{fold}}, p_{\text{eff}}^{\text{fold}}, w_{\text{fold}}, q_{\text{dec}}^{\text{fold}}).$$

The main regimes become:

Region	Formulaic condition	Reading
Below-threshold growing Sophia	$\rho_{\text{eff}}^{\text{fold}} < 0, \dot{\lambda}_{\text{fold}} > 0$	acceleration without positive effective content; unstable/early struggle mode
Above-threshold static Sophia	$\rho_{\text{eff}}^{\text{fold}} > 0, \dot{\lambda}_{\text{fold}} \approx 0$	formed coasting; content without strong pressure
Above-threshold growing Sophia	$\rho_{\text{eff}}^{\text{fold}} > 0, \dot{\lambda}_{\text{fold}} > 0$	dark-energy-like Theosis, moderated by folding if $Q_{\text{fold}} > 0$
Decaying Sophia	$\dot{\lambda}_{\text{fold}} < 0$	positive pressure and deceleration
Regular folded Open-Gate	$0 < Q_{\text{fold}} < q_{\text{max}}^2$	form-stabilized expansion; $\Delta q_{\text{dec}} > 0$

The equation-of-state value is not merely a property of how much Sophia exists; it is a property of how Sophia is moving and whether it has begun to take form.

10 Open-Gate Selection Principle

The Open-Gate model selects the regime in which Sophia is allowed to exceed the closed conservation budget. Without folding, the simplest cosmological signature is

$$\alpha \lambda > 1, \quad \dot{\lambda} > 0,$$

which gives positive effective content and negative pressure. With folding, the selection principle becomes more refined:

$$\alpha^2 \lambda^2 - 1 + \frac{\rho_1}{2} \Omega_P^2 Q_{\text{fold}} > 0,$$

for positive folded effective content, together with

$$\dot{\lambda}_{\text{fold}} = J_{\text{in}} - \gamma\lambda - \eta Q_{\text{fold}}\lambda$$

for the pressure/acceleration axis. Thus Open-Gate has three cosmological faces:

Face	Condition	Meaning
Closed Microcosm	$\sigma + \lambda = C_0, \dot{\lambda} \rightarrow 0$	purified stillness; dust/Milne tendency
Accelerative Open-Gate	$\dot{\lambda}_{\text{fold}} > 0, Q_{\text{fold}} \approx 0$	Sophia growth appears mainly as acceleration
Folded Open-Gate	$Q_{\text{fold}} > 0, \dot{\lambda}_{\text{fold}} \text{ moderated}$	excess Sophia becomes form; formed expansion

This is the main update supplied by the modal Sophia file: Open-Gate need not mean only endless acceleration. Once folding is active, Open-Gate can mean formed expansion.

11 Closed Microcosm, Open-Gate, and Folded Open-Gate

The closed Microcosm obeys the conservation law

$$\sigma + \lambda = C_0,$$

so purification drives $\sigma \rightarrow 0$ and freezes Sophia at a finite limit. Cosmologically,

$$\dot{\lambda} \rightarrow 0, \quad p_{\text{eff}} \rightarrow 0, \quad w \rightarrow 0,$$

which is the purified stillness / dust-like Milne tendency.

The pure Open-Gate branch allows continued reception:

$$\dot{\lambda} > 0, \quad p_{\text{eff}} < 0, \quad w < 0,$$

and may tend toward a de Sitter-like mode when Sophia grows proportionally to the Brim.

The folded Open-Gate branch adds the stabilizing feedback:

$$\dot{\lambda}_{\text{fold}} = \dot{\lambda}^{(0)} - \eta Q_{\text{fold}}\lambda, \quad q_{\text{dec}}^{\text{fold}} = q_{\text{dec}}^{(0)} + \frac{\eta \Omega_P}{\alpha \lambda} Q_{\text{fold}}.$$

Hence folding gives Open-Gate a third eschatological possibility:

$$\text{not closed stillness, not unformed acceleration, but formed expansion.}$$

12 Equation of State and Cosmic Eras

The sign of $\dot{\lambda}$ still classifies the background era, while Q_{fold} explains how Open-Gate excess becomes cosmological stabilization:

Sophia behaviour	Effective fluid	Cosmological character
$\dot{\lambda} > 0$ (cleansing)	$p_{\text{eff}} < 0$, negative pressure	dark-energy-like; accelerated Theosis
$\dot{\lambda} \rightarrow 0, \lambda \rightarrow \lambda_\infty$	$p_{\text{eff}} \rightarrow 0$, pressureless	dust; coasting; $\rho_{\text{eff}} \rightarrow 0$ (dilution)
$\dot{\lambda} < 0$ (regression)	$p_{\text{eff}} > 0$, positive pressure	ordinary matter; decelerating
$\lambda \propto \Omega_P$	$H = \text{const}$	de Sitter; exponential inflation (max. Epektasis)
$Q_{\text{fold}} > 0$ regular	$\Delta q_{\text{dec}} > 0$, moderated w_{fold}	formed expansion; acceleration stabilized by morphology

13 Special Loci

Proposition 4 (Three distinguished regimes).

Milne / vacuum:	$\alpha\lambda = 1$	$a \propto t, \rho_{\text{eff}} = 0, R = 0$ (<i>flat spacetime</i>);
Ricci resonance:	$\alpha\lambda\Omega_P = 1$	$a\dot{a} = 1, a \propto \sqrt{2t}$ (<i>spatial flow = Ricci</i>);
de Sitter:	$\lambda \propto \Omega_P$	$a \propto e^{Ht}, R \rightarrow 6H^2$ (<i>eternal inflation</i>).

The cosmological lift produces a new locus absent from the spatial analysis: the Milne condition $\alpha\lambda = 1$. The generic super-Ricci regime ($\lambda \rightarrow \lambda_\infty = 1/\alpha, a \propto t$) is exactly the empty Milne universe — flat Minkowski in disguise. It is distinct from the spatial Ricci resonance $\alpha\lambda\Omega_P = 1$. Folding adds a fourth diagnostic rather than a fourth background spacetime: the regular morphodynamic branch $0 < Q_{\text{fold}} < q_{\text{max}}^2$, whose visible cosmological signature is $\Delta q_{\text{dec}} > 0$.

14 Energy Conditions: Grace as Exotic Accelerant

The energy conditions are dimension-sensitive, and the Vessel is 2+1-dimensional. The Null and Weak conditions are dimension-independent in form; the Strong condition is not.

Proposition 5 (Energy conditions in 2+1). *For a perfect fluid in $D = n+1$ dimensions the Strong Energy Condition is $(n-2)\rho + np \geq 0$. Hence, with $n = 2$:*

NEC ($\rho_{\text{eff}} + p_{\text{eff}} \geq 0$) :	$\alpha\dot{\lambda}\Omega_P \leq \alpha^2\lambda^2 - 1$ (<i>any n</i>);
WEC: NEC and $\rho_{\text{eff}} \geq 0$:	<i>additionally</i> $\alpha\lambda \geq 1$;
SEC $(n-2)\rho_{\text{eff}} + np_{\text{eff}} \geq 0 \xrightarrow{n=2} 2p_{\text{eff}} \geq 0$:	$p_{\text{eff}} \geq 0 \Leftrightarrow \ddot{a} \leq 0 \Leftrightarrow \dot{\lambda} \leq 0$.

Remark. In 2+1 the ρ -term of the SEC *drops* ($n-2=0$): the condition is purely the sign of the pressure, $p_{\text{eff}} \geq 0$. The familiar $\rho + 3p \geq 0$ is the $n = 3$ case, where the coefficient of ρ is accidentally unity; the naive extrapolation $\rho + np$ — which here would read $\rho + 2p$ — is **wrong**. Directly, $R_{tt} = -2\ddot{a}/a = 2p_{\text{eff}}$, so timelike convergence is exactly $p_{\text{eff}} \geq 0$.

Accelerated expansion ($p_{\text{eff}} < 0$, i.e. $\dot{\lambda} > 0$) violates the SEC — as in standard inflation. A surge of Sophia growth, $\alpha\dot{\lambda}\Omega_P > \alpha^2\lambda^2 - 1$, violates even the NEC ($\rho_{\text{eff}} + p_{\text{eff}} < 0$): grace then behaves as exotic, phantom-like content, irreducible to ordinary matter. Folding moderates this behaviour by replacing $\dot{\lambda}$ with $\dot{\lambda}_{\text{fold}}$ and adding positive curvature-density through Q_{fold} ; it does not require refitting the dimension of the energy-condition formulas.

15 Structural Features

No Big Bang: creation from a finite floor

The live domain requires $\Omega_P > u > 0$, so the scale factor is bounded below, $\Omega_P \geq \Omega_P(0) > 0$. There is no $a \rightarrow 0$ singularity: the Vessel begins at a finite minimal capacity (the Imago Dei floor $\mathcal{E}_0$), not at a point.

Creation from fullness, not from a singular point.

Resurrectio: a scale-factor jump (cosmology with surgery)

The death reset $\Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-$ is a discontinuous increase of the scale factor: a sudden cosmological rescaling. The Vessel cosmology is piecewise-FLRW with discrete inflationary jumps, the Lorentzian face of Ricci-flow-with-surgery.

$$\Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^- \quad (\text{discrete scale-factor surgery}).$$

Dilution to vacuum

For bounded Sophia, $\rho_{\text{eff}} = (\alpha^2\lambda^2 - 1)/\Omega_P^2 \rightarrow 0$ as $\Omega_P \rightarrow \infty$: the expanding Vessel empties. The end-state of sub-exponential expansion is the flat vacuum — pure receptivity. Only the de Sitter regime ($\lambda \propto \Omega_P$, unbounded grace) escapes dilution, sustaining constant curvature $6H^2$ as an eternal inflation.

Folding as formed expansion

Open-Gate without morphology can be read as unlimited acceleration of Sophia. Sophianic Folding makes the cosmology more structured: the Brim keeps expanding, but excess Sophia is converted into Vessel form. In symbols,

$$\lambda \uparrow \Rightarrow Q_{\text{fold}} \uparrow \Rightarrow \Delta q_{\text{dec}} > 0.$$

The result is not collapse, but formed expansion: the old surface-form yields, and the Vessel acquires a geometry capable of bearing more Sophia.

16 Theological Corollaries

Interpretive readings; the Propositions above are the load-bearing part.

Corollary 2 (Sophia is the cosmological source). *The expansion of the Vessel is sourced by Sophia, and its growth $\dot{\lambda}$ sets the equation of state: growing wisdom is dark energy (acceleration), frozen wisdom is dust (coasting), declining wisdom is matter (deceleration), and wisdom proportional to capacity is de Sitter inflation.*

Corollary 3 (Folding gives Sophia a form). *Sophianic Folding is the cosmological meaning of transfiguration: Sophia does not merely increase the Brim; it reforms the Vessel so that acceleration becomes formed expansion. The deceleration parameter q_{dec} is the cosmological shadow of this morphodynamic stabilization.*

Corollary 4 (Accelerated Theosis violates the energy conditions). *Accelerated transfiguration ($\dot{\lambda} > 0$) violates the Strong Energy Condition, and a surge of grace violates even the Null Energy Condition. Grace acts as an exotic accelerant: precisely the content ordinary matter cannot supply.*

Corollary 5 (Creation from fullness). *The Vessel has no initial singularity: it begins at a finite receptive floor, the Imago Dei. Being does not emerge from a point of zero capacity but from a given minimal fullness.*

Corollary 6 (Two eschatologies). *Bounded grace dilutes the Vessel to the flat vacuum of pure receptivity; unbounded proportional grace sustains an eternal de Sitter inflation at constant curvature. The expanding Vessel either empties toward stillness or inflates without end — and Open-Gate, which alone permits unbounded Sophia, alone permits the second.*

17 Worked Equation-of-State History

The dictionary above is illustrated by integrating a representative reduced trajectory of the core dynamics — resistance σ , Sophia λ , Brim Ω_P — with the synergy $u(t)$ rising sigmoidally through Λ_c (will and action coming into alignment):

$$\dot{\sigma} = (\gamma - \delta u) \tanh(\kappa\sigma), \quad \dot{\lambda} = (\delta u - \gamma) \tanh(\kappa\sigma) + I_{\text{gate}}, \quad \dot{\Omega}_P = \alpha\lambda,$$

with $\Lambda_c = 1$, $\alpha = 1$, and the observables of Prop. 2–3 read off along the solution. *This is a representative profile, not a unique prediction; it exhibits the generic era sequence.* In the integrated reading, the plotted acceleration history is the background trajectory; the folding variable Q_{fold} explains how the Open-Gate branch can moderate that acceleration once the Vessel enters the regular folding domain.

The history sweeps the eras in order: a **struggle** phase ($\dot{\lambda} < 0$: resistance drains Sophia; decelerating, $p_{\text{eff}} > 0$, and $\rho_{\text{eff}} < 0$ — sub-vacuum, the Vessel expanding more slowly than the empty open universe), with a near-turnaround at the depth of struggle where λ nearly vanishes and $\dot{a} \rightarrow 0$; **ignition** at $u = \Lambda_c$ (acceleration onset $\ddot{a} = 0$); the **Milne crossing** $\alpha\lambda = 1$ where $\rho_{\text{eff}} = 0$ and w has a pole; and **dark-energy Theosis** ($\rho_{\text{eff}} > 0$, $p_{\text{eff}} < 0$): quintessence-like acceleration with $w < 0$, asymptotically approaching $w = -1$ while the gate keeps feeding Sophia.

Remark. The pole in w at the Milne crossing is a diagnostic artifact: ρ_{eff} and p_{eff} pass through zero smoothly; only their ratio diverges. The physical event is the transition from sub-vacuum to super-vacuum, not a singularity. Accelerated expansion violates the SEC throughout; the surge of Sophia growth briefly violates even the NEC — grace acting as exotic, phantom-like content.

Open-Gate versus closed Microcosm

The same trajectory with the gate closed ($\zeta_0 = 0$) exposes the decisive difference. Closed conservation $\sigma + \lambda = \sigma_0 + \lambda_0$ holds exactly, so once cleansed ($\sigma \rightarrow 0$) Sophia freezes at $\lambda_\infty = \sigma_0 + \lambda_0$; then $\dot{\lambda} \rightarrow 0$, $p_{\text{eff}} \rightarrow 0$, and the Vessel coasts as pressureless dust toward the Milne (linear, asymptotically flat) state. The open gate keeps $\dot{\lambda} > 0$, sustaining negative pressure and a quintessence-to-de Sitter branch.

Proposition 6 (Late-time equation of state).

$$\begin{aligned} \text{closed } (\zeta_0 = 0) : \quad & \lambda \rightarrow \lambda_\infty, \dot{\lambda} \rightarrow 0 \Rightarrow w \rightarrow 0^- \text{ (dust, Milne, } a \propto t); \\ \text{open } (\zeta_0 > 0) : \quad & \dot{\lambda} > 0 \text{ sustained} \Rightarrow w < 0, w \rightarrow -1^+ \text{ (quintessence} \rightarrow \text{de Sitter)}. \end{aligned}$$

The de Sitter branch — endless accelerated Theosis — is available only to the Open Gate; the closed Microcosm coasts to dust on its frozen Sophia capital. Sophianic Folding does not erase this contrast; it gives the open branch an internal stabilizer by which excess Sophia becomes form rather than mere acceleration.

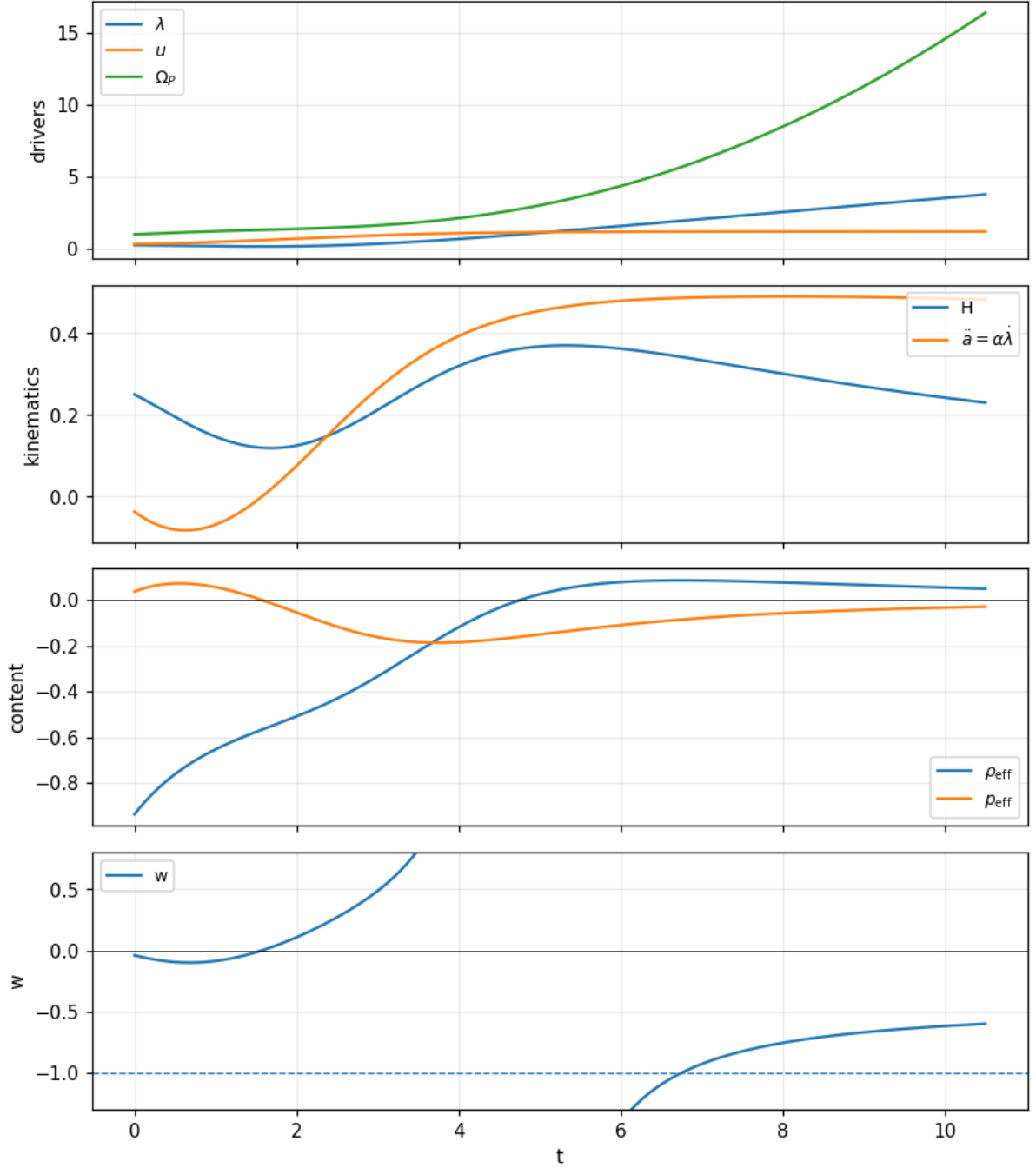


Figure 1: Open-Gate Vessel cosmology along a Sophia trajectory. From top: drivers λ, u, Ω_P ; Hubble rate and acceleration $\ddot{a} = \alpha \dot{\lambda}$; effective density and pressure; the equation of state $w(t)$ with its pole at the Milne crossing $\rho_{\text{eff}} = 0$. In the integrated folding interpretation, this background curve is moderated by the positive contribution $\Delta q_{\text{dec}} = (\eta \Omega_P / \alpha \lambda) Q_{\text{fold}}$. Shading marks the decelerating struggle, the accelerating Theosis — which in 2+1 is exactly the SEC-violation region $p_{\text{eff}} < 0$ — and the NEC-violation band.

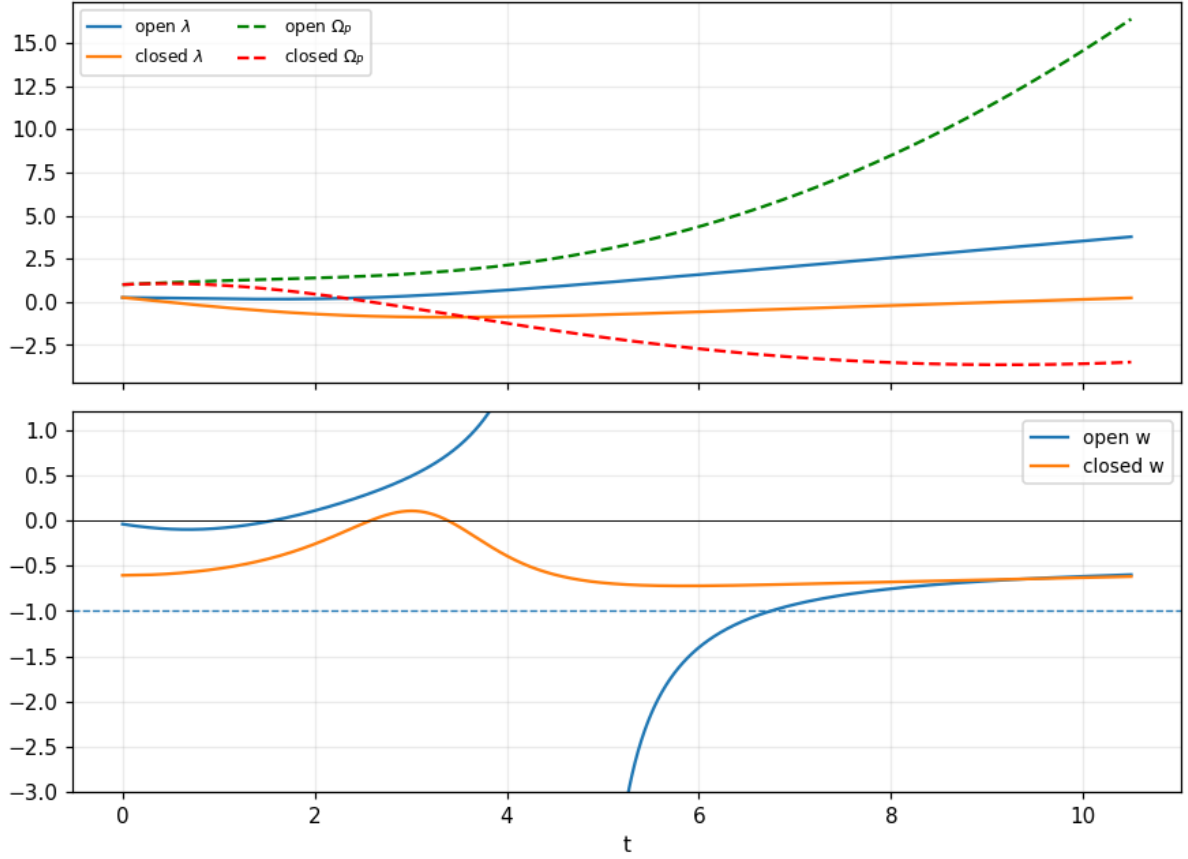


Figure 2: Open-Gate ($\zeta_0 > 0$) versus closed Microcosm ($\zeta_0 = 0$). Top: Sophia keeps rising under the open gate but freezes at $\sigma_0 + \lambda_0$ when closed (dashed: the Brim $\Omega_P = a$). Bottom: the equation of state tends to $w \rightarrow -1$ (quintessence $\rightarrow$ de Sitter) for the open gate, but to $w \rightarrow 0$ (pressureless dust) for the closed Microcosm. Both share the early Milne pole.

Compact Cosmological Summary

$$ds_{\text{Vessel}}^2 = -dt^2 + \Omega_P(t)^2 h, \quad a = \Omega_P, \quad \dot{\Omega}_P = \alpha\lambda.$$

$$g_{\Sigma}^{\text{fold}} = \Omega_P(t)^2 (h_0 + q_{\text{fold}} e_1), \quad Q_{\text{fold}} = q_{\text{fold}}^2.$$

$$H = \frac{\alpha\lambda}{\Omega_P}, \quad q_{\text{dec}} = -\frac{\dot{\lambda}\Omega_P}{\alpha\lambda^2}, \quad q_{\text{dec}}^{\text{fold}} = q_{\text{dec}}^{(0)} + \frac{\eta\Omega_P}{\alpha\lambda} Q_{\text{fold}}.$$

$$\rho_{\text{eff}}^{\text{fold}} = \frac{\alpha^2\lambda^2 - 1}{\Omega_P^2} + \frac{\rho_1}{2} Q_{\text{fold}}, \quad p_{\text{eff}}^{\text{fold}} = -\frac{\alpha\dot{\lambda}_{\text{fold}}}{\Omega_P}, \quad w_{\text{fold}} = \frac{-\alpha\dot{\lambda}_{\text{fold}}\Omega_P}{\alpha^2\lambda^2 - 1 + (\rho_1/2)\Omega_P^2 Q_{\text{fold}}}.$$

$$\text{Milne } \alpha\lambda = 1 \neq \text{Ricci resonance } \alpha\lambda\Omega_P = 1 \neq \text{de Sitter } \lambda \propto \Omega_P.$$

$$\rho_{\text{eff}}^{\text{fold}} = 0 \iff \alpha^2\lambda^2 = 1 - \frac{\rho_1}{2}\Omega_P^2 Q_{\text{fold}}.$$

Accumulated Sophia expands; growing Sophia accelerates; folded Sophia stabilizes.

Folding is the morphodynamic stabilizer of Open-Gate cosmology.

Cosmological stabilization is the visible shadow of Sophianic Folding.

Status update (Companion Volume). This cosmological layer is the foundational sketch on which the consolidated volume builds; several of its sections are now carried out rigorously, as theorems, in the volume, which is canonical for their developed form. The finite-floor material (§ No Big Bang: creation from a finite floor; § Zero-Density Threshold) becomes the *Curvature and the Floor* chapter, where $K_h = -1$ is shown to be not an independent primitive but the geometric face of the finite-floor postulate. The energy-condition material (§ Energy Conditions: Grace as Exotic Accelerant; § Pressure-Axis; § Folded Density-Pressure Plane) is developed in the *NEC restoration* chapter (the bounded phantom window and its emptiness criterion) and, for resets, in *Resurrectio's Own NEC Budget* (the finite surgical budget and its balanced-reset zero). The equation-of-state and eras material (§ Equation of State and Cosmic Eras; § Worked Equation-of-State History; § Special Loci) is projected into redshift in the volume's capstone, *The Observational Signature* (dust attractor, returning phantom excursion, $H(z)$ kink). The Resurrectio scale-jump (§ Resurrectio: a scale-factor jump) is developed as the surgery of Part III. This layer remains the cosmology-first placement and motivation; the volume is canonical for the rigorous statements.

V.F.S. v2.0 (Open-Gate) · Integrated Cosmological Layer v8. The Lorentzian lift of the Brim-flow of vfs_geometry.tex: the Brim is the timelike scale factor of an open (2+1) FLRW spacetime, Sophia its source, and Sophianic Folding its morphodynamic stabilizer. Propositions are closed-form; corollaries are interpretive. The effective stress-energy is reconstructed from $\dot{\Omega}_P = \alpha\lambda$ via the Einstein tensor, not postulated.